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Writing the Conclusion

Isti Hidayati (isti.hidayati@ugm.ac.id)

Teaching Team of Ethics and Scientific Writing Technique Course

Faculty of Engineering
Universitas Gadjah Mada, 2025

The Shape of a Research Article

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TODAY OFFER

Prepare Figures/Graphs

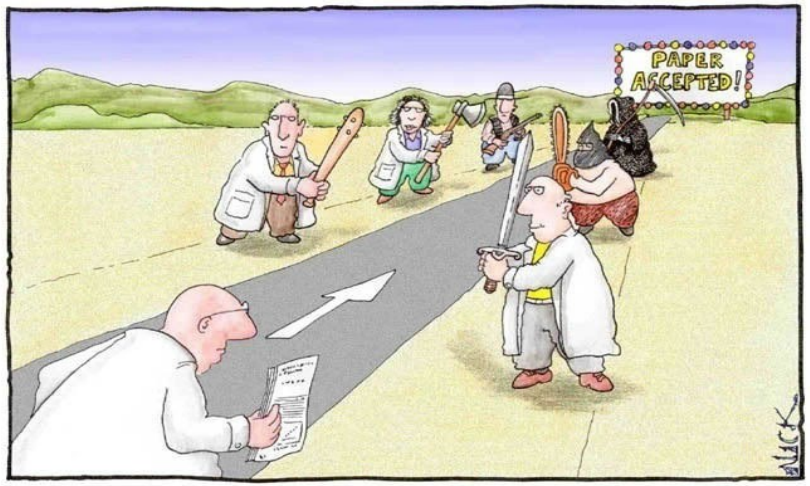
Prepare Results/Discussion

Prepare Methodology

Prepare Introduction

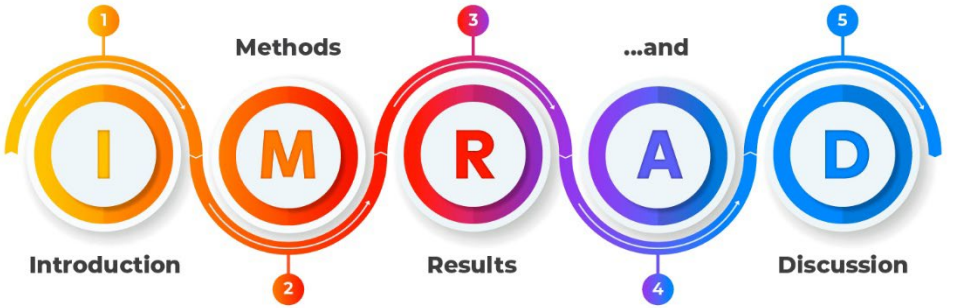
Prepare Conclusion, Abstract & Title

Order of our work when we prepare our manuscript

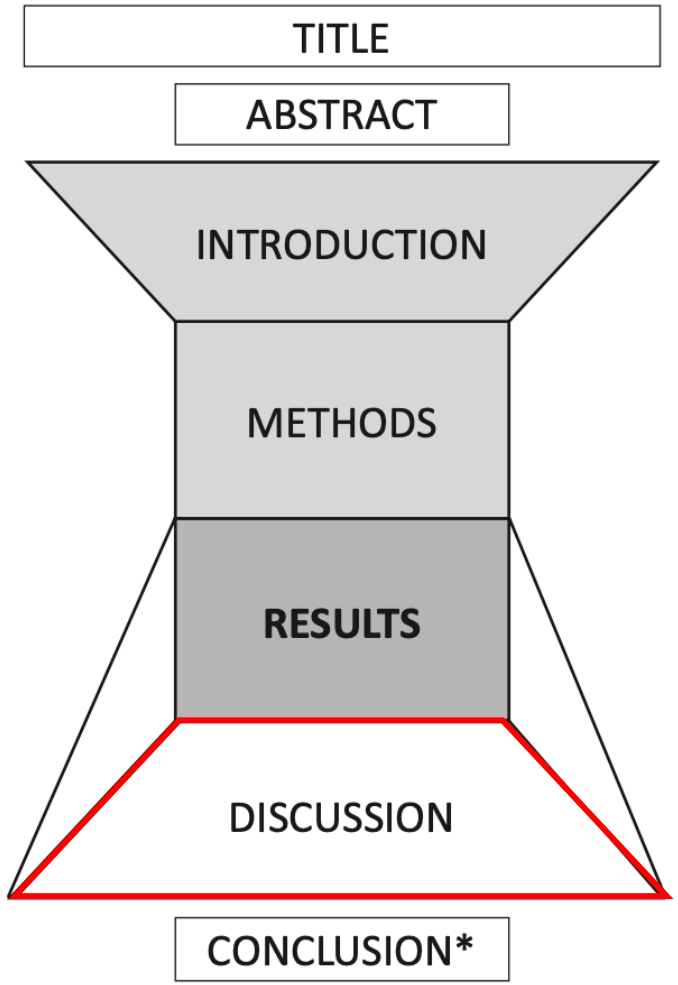


Most scientists regarded the new streamlined peer-review process as 'quite an improvement.'

www.weirdscience.ca



The Shape of a Research Article



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Contents

1. Background on Conclusion
2. General contents in a conclusion
3. Exercise

Background on Conclusion

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- Mostly stands alone as a separate section. However, in some journals, conclusion can be part of the Discussion.
- On average 100-200 words or 2 paragraphs; some journals prefer bullet points, while there are journals that prefer a longer paragraph → check the style of your target journal
- Content overlaps with abstract, introduction, and discussion
- Conclusion goes beyond summary → take-home message, focus on outcome and impacts of your research
- Often use simple present and present perfect tense
- *May* be reversed from the abstract → from specifics to generalization
- Be concise!

Background on Conclusion

SUBTITLES Option 1	SUBTITLES Option 2	SUBTITLES Option 3	SUBTITLES Option 4
Results	Results	Results and Discussion	Results
Discussion	Discussion		Ø
Conclusion/s	Ø	Conclusion/s	Conclusion/s
There is a separate Conclusion section of 1-2 short paragraphs	*The last 1-2 paragraphs of the Discussion section are similar in form and content to a Conclusion section*	*There is a separate Conclusion section of 1-2 short paragraphs*	*There may be a long Conclusion section that is initially similar in form and content to a Discussion section and which ends with 1-2 paragraphs that are similar in form and content to a Conclusion section*

General contents in a conclusion

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1. What the study is about
2. What the study has achieved
3. Relevant background
4. Study gap/aim/importance
5. The method/approach
6. Key results with evaluative comments
7. Implications of the results
8. Potential or actual limitations → (in some journals, they may also ask for a separate sub section)
9. Potential or actual applications
10. How the study advances knowledge→ contribution to the literature or practice
11. Future research directions → can also be policy insights → be explicit and avoid ambiguity

DO NOT copy paste from your previous section, make efforts to paraphrase/restate the sentences.

Conclusion paragraph - example

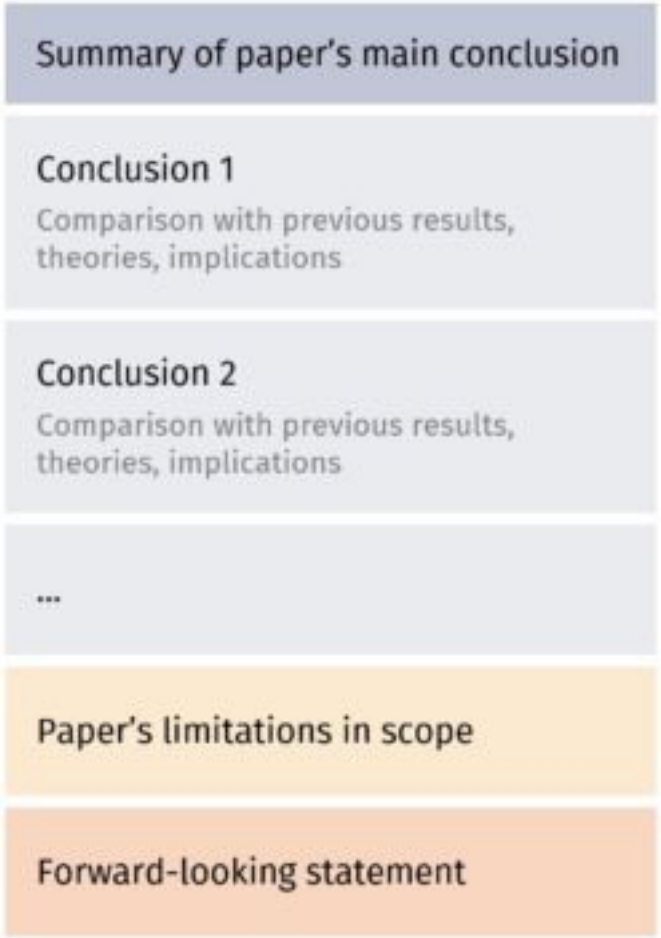
Media can have both positive and negative influences on children. Media, particularly television that is of high quality, expose children to learning opportunities and can provide appropriate role models for children. Early exposure to age-appropriate programs designed around an eduational curriculum are found to enhance cognition and academic performance. Research suggests that parents who select well-designed, age-appropriate programs and view these programs with their children maximise the positive effects of media on their children. However, media which shows excessive or gratuitous violence can have long term negative effects on a child's behaviour, cognition, and physiological arousal. There is also evidence to suggest that over-exposure to media can damage some children's ability to connect with real people in everyday situations and there appears to be a strong link between over-exposure to media and the rise of obesity in children. Importantly, it appears that the risk of harmful effects of media are significantly reduced when parents monitor both the amount of time children spend in front of the television or computer screen as well as the content and suitability of the programs. Therefore, children can be both positively and negatively influenced by media, and the best safe guard against exposing children to harmful effects of media is by careful monitoring of the quantity and quality of media by parents.

CONCLUSION

- Restate thesis
- Summary of main points (General conclusions)
- Final comment (Draw together question, evidence and conclusion)

Conclusion models

Source: https://libguides.newcastle.edu.au/how-to-write-an-essay/conclusion



Conclusion models

Source: https://mitcommlab.mit.edu/nse/commkit/journal-article-discussion/

Physical cryptographic verification of nuclear warheads

R. Scott Kemp[a,1,2], Areg Danagoulian[a,1], Ruaridh R. Macdonald[a,1], and Jayson R. Vavrek[a,1]

[a]Laboratory for Nuclear Security and Policy, Massachusetts Institute of Technology, Cambridge, MA 02139

Kemp, et. al. PNAS (2016)
doi: 10.1073/pnas.1603916113

Overall summary and importance statement

Definining the scope of the problem considered

Trends in nuclear disarmament indicate a need for a provably sound, complete, and information-secure method for authenticating nuclear warheads. The system proposed here seeks to meet these requirements with a mathematically quantifiable physical framework and logically sound set of axioms. By the virtue of being in the public domain, various aspects of the system can be freely studied and scrutinized by any participant in the arms reduction process. In addition to theoretical considerations, there may be important operational limitations to implementing such a system in existing national-security environments, which will require further study by government experts.

Soundness against strategically meaningful hoax warheads is achieved by careful attention to the logical connections that link observables to conclusions about warhead authenticity; by exploiting a process in nuclear physics that acts as a naturally occurring one-to-one function between energy and isotope identity; and through the introduction of a single-pixel tomographic transform that provides a unique measure of the warhead's geometry. Information is protected by the intrinsic physics of the system rather than relying on difficult-to-certify electronic circuits or software. Although electronics are used to operate the detectors and to measure signals leaving the shielded area, no sensitive information ever becomes experimentally observable.

Simulations for a bremsstrahlung-based system using available technologies and the simulated geometries suggest that two 21-s measurements should be able to reject a canonical set of hoaxes with greater than 99.9% probability, while falsely alarming on an authentic warhead about once in every 10,000 warheads. Realistic foil and warhead geometries will almost certainly require somewhat longer measurements than those simulated. Work remains to validate these simulations in an experimental setting. NRF cross-section data for some isotopes, mainly those not used here, are still sparse and may need to be measured if those isotopes are to be included in the verification process.

If a bremsstrahlung beam is used, several information security questions remain open. The information content of the continuum underneath the NRF signal is at present poorly understood; the simultaneous observation of multiple NRF transitions, which is unavoidable with bremsstrahlung, will require careful study of countermeasures to ensure the system remains underconstrained; and dose to the warhead will be high when using bremsstrahlung. Many of these problems can be avoided by using a tunable monochromatic photon source.

Finally, although the example system suggests excellent performance for canonical hoaxes and measurements on the order of minutes, a relationship between measurement time and the upper bound on the probability of failing to reject any possible hoax would be a useful future development. As these probabilities are inherently a function of hoax geometry, such a statement requires a description of the space of all possible hoax objects, which might first require the development of an inversion formula for the $\mathcal{K}$ transform. This remains as future work.

Engineering and physics considerations that make this solution unique

Importance statement about engineering solution

Statement of specific results

Identification of future data necessary for implementable system

Limitations on a proposed extension identified

Potential solution to identified limitation

Work to be continued in the future

Source: http://mitcommlab.mit.edu/nse/wp-content/uploads/sites/4/2019/10/JournalDiscussion-Kemp.jpg

Exercise

Referring to the 11 elements (slide 3 on the general contents in conclusion) can you identify them in this sample manuscript?

The oxidative corrosion of carbide inclusions at the surface of uranium metal during exposure to water vapour

Highlights

High resolution imagery (FIB, SEM and SIMS) of carbide inclusions in uranium metal. ► Real time images following the reaction of the carbide inclusions with water vapour. ► Shown preferential consumption of carbide over that of the bulk metal. ► Quantity of impurities in the metal therefore seriously influence reaction rate. ► Metal purity must be considered when storing uranium in air or moist conditions.

Abstract

The reaction between uranium and water vapour has been well investigated, however discrepancies exist between the described kinetic laws, pressure dependence of the reaction rate constant and activation energies. Here this problem is looked at by examining the influence of impurities in the form of carbide inclusions on the reaction. Samples of uranium containing 600 ppm carbon were analysed during and after exposure to water vapour at 19 mbar pressure, in an environmental scanning electron microscope (ESEM) system. After water exposure, samples were analysed using secondary ion mass spectrometry (SIMS), focused ion beam (FIB) imaging and sectioning and transmission electron microscopy (TEM) with X-ray diffraction (micro-XRD). The results of the current study indicate that carbide particles on the surface of uranium readily react with water vapour to form voluminous $UO_3 \cdot xH_2O$ growths at rates significantly faster than that of the metal. The observation may also have implications for previous experimental studies of uranium–water interactions, where the presence of differing levels of undetected carbide may partly account for the discrepancies observed between datasets.

Exercise

Referring to the 11 elements (slide 3 on the general contents in conclusion) can you identify them in this sample manuscript?

The oxidative corrosion of carbide inclusions at the surface of uranium metal during exposure to water vapour

Conclusions

Examination of cast α-uranium surfaces after exposure to water vapour in an ESEM instrument at 19 mbar and 20 °C indicated surface corrosion had occurred, with secondary growths determined to be $UO_3 \cdot H_2O$ (metaschoepite) forming at carbide inclusions across the surface. Over the period of a week the growths were observed to increase in size, consuming only the carbide particles and not the surrounding metal. In some cases complete decomposition of the surface carbides was observed.

From the results of the current study it is apparent that the carbide particles reacted more readily with the water vapour than the metal. Resultantly it is suggested that disparities between previous studies of the uranium–water reaction may be attributable to differential purities of uranium metal used by different research groups, with resultantly different populations of carbide particles.

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Notice that the **abstract** goes from general to detail

While **conclusion** goes from detail to general

Abstract

The reaction between uranium and water vapour has been well investigated, however discrepancies exist between the described kinetic laws, pressure dependence of the reaction rate constant and activation energies. Here this problem is looked at by examining the influence of impurities in the form of carbide inclusions on the reaction. Samples of uranium containing 600 ppm carbon were analysed during and after exposure to water vapour at 19 mbar pressure, in an environmental scanning electron microscope (ESEM) system. After water exposure, samples were analysed using secondary ion mass spectrometry (SIMS), focused ion beam (FIB) imaging and sectioning and transmission electron microscopy (TEM) with X-ray diffraction (micro-XRD). The results of the current study indicate that carbide particles on the surface of uranium readily react with water vapour to form voluminous $UO_3 \cdot xH_2O$ growths at rates significantly faster than that of the metal. The observation may also have implications for previous experimental studies of uranium–water interactions, where the presence of differing levels of undetected carbide may partly account for the discrepancies observed between datasets.

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Conclusion

To describe **what is in the paper**, **what is implied by the results**, and **what has been achieved** in the study

Present simple OR present perfect

- ***We have synthesized and characterized*** *two fluorescent rhodamine-containing PTP probes…*
- ***It is argued that*** *this rate law is consistent with reaction mechanisms based on recent observations…*
- *This long-standing problem* ***is solved*** *by combining micron-sized and nano-sized tougheners in a synergistic manner.*
- *The rate order of the water oxidation reaction* ***has been investigated*** *on hematite photoanodes by…*
- *In the present paper, some shortcomings of the concept of cross-section classification* ***have been highlighted****, and, as an alternative treatment, the continuous strength method* ***has been introduced****.*
- *The level of enhancement in resistance offered by the continuous strength method over conventional design methods for steel structures* ***has been found to be*** *approximately…*
- *Finally,* ***it is shown that*** *the fluorescent probes are capable of monitoring…*

Past simple to repeat key aspects/methods/results

- *The phase behavior of (CO_2 + H_2O)* ***was measured*** *at temperature from 298.15 K to 448.15 K,…*
- *Damage accumulation* ***evolved*** *slowly at lower strains, but* ***continued*** *at a rapid rate immediately prior to fracture, which is a typical ductile fracture characterized by dimples.*
- *In some cases complete decomposition of the surface carbides* ***was observed****.*
- *The system* ***was found to*** *induce a well-localized pattern of…*

Conclusion

Owning your **contribution**.

Writer's bias → writer feels familiar with the topic, while reader often jumps to Conclusion.

***Kinetic analyses suggest** that these probes are active site directed and inactivate a broad range of PTPs in a time- and concentration-dependent fashion. **Direct in-gel fluorescence scanning indicates** that the fluorescent probes form a covalent adduct with the PTPs.*

Lack of relevant background
Reader may feel 'lost'

How to improve:
- Use past simple or present perfect
- Use active rather than passive voice
- Use 'human' grammatical object
- First sentence should explicitly refers to the paper

Conclusion

Owning your **contribution**.

Now compare the writing style between these sentences:

Kinetic analyses suggest *that these probes are active site directed and inactivate a broad range of PTPs in a time- and concentration-dependent fashion.* ***Direct in-gel fluorescence scanning indicates*** *that the fluorescent probes form a covalent adduct with the PTPs.*

- **We have synthesized and characterized** two fluorescent rhodamine-containing PTP probes that are highly sensitive for direct in-gel visualization of PTP activity.
- Examination of cast α-uranium surfaces after exposure to water vapour in an ESEM instrument at 19 mbar and 20 °C **indicated that** surface corrosion had occurred,…
- This transition **was confirmed by** agreement between three different analyses.
- **Finally, it is shown here that** the fluorescent probes are capable of monitoring H_2O_2-mediated PTP inactivation,…

- **This study clearly demonstrates** the ability of a metal oxide semiconductor surface, such as hematite, to drive a multihole reaction,…
- **In the present paper**, some shortcomings of the concept of cross-section classification **have been highlighted**, and, as an alternative treatment, the continuous strength method **has been introduced.**
- **From the results of the current study** it is apparent that the carbide particles reacted more readily with the water vapour than the metal.

Discussions?

- Can you identify key take aways from the conclusion?
- As fellow reader, what turn you off the most when reading the conclusion?
- Have you ever read conclusion of an article that left you confused? What makes you confused?

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Thank you